

Program: Diploma in Printing Technology	
Course Code: 6109B	Course Title: Printing & Waste Management Lab
Semester: 6	Credits : 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To develop functional experience with common printing practice.
- To build up hand on experience to do quality control with printing.
- To develop precise skill to perform common waste management practice in printing.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Sheef-fed offset		Mechanism of Printing Machines I	III
Printing with Sheet-fed offset.		Advanced Printing Machines Workshop I, II &III.	III, IV, V

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	To perform common printing practice.	14	Applying
CO2	To perform quality control with printing.	13	Applying
CO3	To perform common waste management practice in printing.	14	Applying
	Lab Exam	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3	3		3			
CO3	3			3	3	2	2

3-Strongly mapped**2-Moderately mapped****1-Weakly mapped****Course Outline:**

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	To perform common printing practice.		
M1.01	Common Prepress practices- Image generation, Image Processing, Proofs, Image carrier preparation of a catalog.	5	Applying
M1.02	Press practices –printing of catalog with sheet fed offset.	5	Applying
M1.03	Post press practices – Cutting, stitching, Trimming, and eyeleting practices for the print finishing of catalog.	4	Applying
CO2	To perform quality control with printing.		
M2.01	Color measurement with standard solid and tint test patches & control.	5	Applying
M2.02	Register measurement and prepare control chart.	4	Applying
M2.03	Density measurement and prepare control chart.	4	Applying
	Lab exam I	2	

CO3	To perform common waste management practice in printing.		
M3.01	Familiarization of Industrial Emission guidelines.	1	Applying
M3.02	Waste Segregation, Disposal, and Reuse.	5	Applying
M3.03	Solvent Recovery of plate developing solution.	5	Applying
M3.04	Formulate an eco friendly Ink.	3	Applying
	Lab exam II	2	Applying

Text / Reference:

T/R	Book Title/Author
T1	Michael Bernard, John Peacock., Handbook of printing and production , Springer Science & BusinessMedia
R1	P. Kipphan, Handbook of Print Media , Springer, 2002.
R2	R.H. Leach and R.J. Pierce Ed, The Printing Ink Manual, 5th Edition , Kluwer Academic publish.
R3	Dejdas LP and Destree TM, Sheet fed offset press operating . GATF, USA
R4	N.R.Elred& T. Scarlet, Chemistry for the Graphic Arts , GATF, 1992.
R5	TulikaDas, Chemistry in Printing, 2nd Edition , BarnanaPrakashani, 2011.
R6	Implementing quality management in graphic arts , HL Apfelberg and MJ Apfelberg.

OnlineResources:

SI No	Website Link
1	https://www.britannica.com/technology/offset-printing
2	https://medium.muz.li/a-beginners-guide-to-screen-printing-by-a-complete-beginner-c5bd393d9238
3	https://en.wikipedia.org/wiki/Control_chart
4	https://www.conserve-energy-future.com/sustainable-practices-waste-management.php
5	https://en.wikipedia.org/wiki/Waste_management .